

Chapter 11

Crack Spread Trading

Trading the refining margin — the spread between crude and the products it becomes

Trading the margin, not the barrel

3-2-1

the benchmark crack

The 3-2-1 crack spread — three barrels of crude yielding two of gasoline and one of distillate — is the standard proxy for a refiner's margin. A trader can buy or sell that margin directly, without ever owning a refinery or a barrel.

A spread, not a flat price — The crack is the difference between product prices and crude. Trading it isolates the refining margin from the direction of oil itself.

Why it exists — It lets refiners lock in profit and lets traders bet on whether refining will be lucrative or thin — independent of where crude goes.

The bridge between chapters — Crack-spread trading joins the crude market of earlier chapters to the product markets — the two priced against each other.

What a crack spread is

The refiner's margin — A crack spread is the gap between what a refiner pays for crude and what it earns selling the refined products — its theoretical gross margin.

The name — To 'crack' crude is to break it into lighter products. The spread measures the value created by that transformation.

Quoted per barrel — It is expressed in dollars per barrel. A positive crack means refining is profitable; a negative crack means products are worth less than the crude.

A market in its own right — Because it can be bought and sold as a single position, the crack spread is a tradable instrument, not just an accounting figure.

The 3-2-1 and its cousins

The 3-2-1 — The standard benchmark: three barrels of crude produce two of gasoline and one of distillate, roughly matching a typical refinery's yield.

The 5-3-2 — A finer approximation — five crude, three gasoline, two distillate — that some refiners use to mirror their slate more closely.

The 1-1-1 and single cracks — Simpler ratios that isolate one product: a gasoline crack or a heating-oil crack against a single barrel of crude.

Convert to one unit — Products are quoted in cents per gallon, so each is multiplied by 42 gallons to reach dollars per barrel before the spread is calculated.

How the 3-2-1 is calculated

Step one — convert to dollars per barrel — Products are quoted per gallon, so multiply by 42. Gasoline at \$2.50/gal becomes \$105/bbl; distillate at \$3.00/gal becomes \$126/bbl.

Step two — value the products — Two barrels of gasoline (\$210) plus one of distillate (\$126) gives a combined product value of \$336.

Step three — subtract the crude — Subtract three barrels of crude. At \$90/bbl that is \$270, so the gross margin is $\$336 - \$270 = \$66$.

Step four — divide by three — Divide by the three barrels processed: $\$66 \div 3 = \22 per barrel. A solid crack, just above the typical \$10-20 range.

Reading the crack — what the number means

Below zero — refining loses money — Products are worth less than the crude. Refiners cut runs or shut units rather than process at a loss, which then tightens product supply.

\$10-20 — normal and healthy — The typical operating band. Refineries run steadily and earn a workable margin; nothing unusual is happening in the market.

\$30-40 — tight market — A disruption, an outage, or strong seasonal demand. Products are outrunning crude, so refiners run flat out to capture the wide margin.

\$80+ — extreme squeeze — A genuine product shortage rationed by price. The ~\$83 diesel crack of 2022 marks the historic peak — rare, violent, and short-lived.

Buying and selling the crack

Selling the crack — The refiner's natural trade: sell product futures and buy crude futures. This locks in the margin and is what a refiner does to hedge.

Buying the crack — The opposite position: buy products and sell crude, profiting if the refining margin widens. Used by traders with a bullish margin view.

One position, three legs — A 3-2-1 crack is executed across multiple contracts at once, but it is managed as a single spread with a single margin figure.

Exchange-listed cracks — Exchanges also list ready-made crack-spread contracts, so the whole position can be traded in one click rather than leg by leg.

Why refiners hedge the crack

Their real exposure — A refiner does not care about the flat price of crude as much as the gap between crude and products — that gap is its actual profit.

Locking in profit — By selling the crack, a refiner fixes its margin for months ahead, protecting against the risk that products fall or crude jumps.

Timing the buy and sell — Refiners buy crude in one month and sell products later, so the crack also hedges the time gap between input and output.

Run decisions — When the crack is wide, refiners run hard and hedge it in; when it turns negative, they cut runs rather than refine at a loss.

What moves the crack spread

The price lag — Products follow crude with a delay. After a crude spike, refiners selling at new product prices briefly capture a wider crack.

Refining capacity — Outages, run cuts, and seasonal maintenance tighten product supply and widen cracks; new capacity narrows them.

Product demand & season — Strong gasoline demand in summer or diesel demand in winter lifts the relevant crack independently of crude.

Feedstock advantage — A refiner buying cheaper crude relative to its products — for example WTI versus Brent-priced fuels — earns a structurally wider crack.

When the crack blows out

\$83

record diesel crack, 2022

When Europe lost Russian diesel in 2022, the diesel crack spiked to around \$83 a barrel — the historic peak. A normal, healthy 3-2-1 crack sits near \$10-20; disruptions can push it past \$30-40, and a true squeeze far beyond.

The 2026 episode — The Hormuz disruption widened cracks again as gasoline and distillate prices outran crude, with the gasoline crack running toward \$28 a barrel.

The WTI edge — US Gulf Coast refiners bought cheaper WTI while selling products priced off Brent, turning the widening WTI-Brent gap straight into margin.

The lesson — The crack can move violently and independently of flat price — which is exactly why it is hedged and traded on its own.

Trading the crack as a strategy

A view on refining, not oil — A crack trade lets a speculator profit from a tightening or loosening product market without taking a flat-price bet on crude.

Lower directional risk — Because both legs move with oil, much of the flat-price risk cancels out, leaving a cleaner bet on the margin itself.

Seasonal and event plays — Traders position for summer gasoline strength, winter diesel demand, or a refinery outage — all expressed through the crack.

Liquidity and the legs — Cracks are most reliable where both the crude and the product contracts are deeply liquid, keeping execution tight.

The risks of crack trading

Not a perfect hedge — The 3-2-1 only approximates a real refinery's slate. The actual margin can drift from the benchmark — this gap is basis risk.

Legging risk — Executing the crude and product legs separately leaves a brief window where prices can move against you before the spread is complete.

Correlation can break — In a shock, products and crude can decouple sharply. That is the opportunity in crack trading, but also its danger.

Still leveraged — A crack is a spread, but each leg is a leveraged futures position. Disciplined margin and risk management remain essential.

Conclusion — the margin made tradable

The crack is the refiner's world — It captures the single number that decides whether refining is profitable — the spread between crude and products.

A hedge and a bet — Refiners sell it to lock in margin; traders buy and sell it to express a view on refining, with crude direction stripped out.

Crude and products, joined — Crack-spread trading is the payoff of the course so far: the crude markets meet the product markets, traded through the futures and spread mechanics covered earlier.

What to remember — Learn to read and trade the crack and you can isolate refining value itself — one of the most powerful spread ideas in the entire oil market.